

SDA Quiz

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Q1 = 4+1 view model consist of 4 ~~process~~ views that are logical view, process view, development view, deployment and plus one view which is use case view.

Logical view

shows objects and their interactions. Stakeholders associated with ~~use case view~~ are enduser, analysts and designers. Functional requirements are considered in this view and UML diagram associated is class diagram. Object?

Process view

shows process and workflows. Stakeholders associated are developers and QA. Non-Functional requirements are considered in this view and UML diagram associated is sequence, Activity diagram. U

Development view

shows describes the building block view of the system. Stakeholders associated are developers.

~~Non-Functional requirements are considered.~~

UML diagram associated is component diagram.

Deployment view

shows how the blocks, components are ^{installed, configured, deployed.} ~~integrated / deployed~~. Stakeholders associated are someone from all stakeholder whose have job is to configure / installation. ~~It~~ UML diagram is deployment diagram. ?

~~It is called 4+1 view model because it consists of four different view model plus a use case / scenario view~~

use case view

shows completed ~~def~~ design by verifying usecases.
It considers different scenarios to check system at runtime.
stake holders associated are endusers and UML diagram
is usecase diagram.

It is called U+1 view model because it consist of four different views ~~to~~ plus an additional usecase / scenario view which varifies or checks each of the four use cases with a given scenario to ensure user requirements are satisfied.

Q2 Characteristics of microservice Architecture

- Each service can be separately deployed, allowing multiple deployments
- Each service has its own database, so multiple databases are used
- Each module can be built using different frameworks.
- APIs are used in between services
- Totally distributed architecture
- Loosely coupled
- ~~Flexible~~ Multiple codebases are used. All the code is not located at a single place

Example 1

Netflix use microservice architecture for recommendations, streaming and other features, so that even if an error occurs in any feature / service whole application doesnot need to be shutdown or ~~crashes~~ doesnot crashes.

Example 2

~~Uber~~ Uber adapted microservice architecture after using monolithic, for so ~~each service can be~~ the application consists of independently deployed units that are more flexible, easy to maintain and test.

Q3

Technical Requirements

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- Application should enable citizen to ^{input} ~~enter~~ their current location so.
- Application should be able to access current location of citizens.
- Application should ask citizens to enter their contact info
- Whenever there is an emergency, there should be an alert button/feature that when clicked by user, alerts the rescue team, ~~hospital~~ and nearby hospitals.
- ~~Application~~ Application should register different users under different categories that are: citizens, rescue team, hospital administration and government authorities.
- There should be a payment gateway through which government authorities could pay the related rescue team, hospital after an emergency.
- There should be a feature through which government authorities could manage drones, like their availability.

